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Table S1. ARIA frequencies through 24 weeks

Category, n (%)	Standard (N=207)	Modified titration (N=212)	Dose skipping (N=210)	Cmax (N=213)
ARIA-E[a,b]	49 (23.7)	29 (13.7)	39 (18.6)	39 (18.3)
Asymptomatic[a,b]	39 (18.8)	23 (10.8)	31 (14.8)	28 (13.1)
Symptomatic[a,b]	10 (4.8)	6 (2.8)	8 (3.8)	11 (5.2)
SAE of ARIA-E[c]	0 (0)	0 (0)	0 (0)	1 (0.5)
ARIA-oedema/effusion[c]	0 (0)	0 (0)	0 (0)	1 (0.5)
Brain oedema[c]	0 (0)	0 (0)	0 (0)	0 (0)
Vasogenic cerebral oedema[c]	0 (0)	0 (0)	0 (0)	0 (0)
ARIA-H[a,d]	52 (25.1)	43 (20.3)	48 (22.9)	44 (20.7)
Asymptomatic[a,d]	52 (25.1)	42 (19.8)	48 (22.9)	40 (18.8)
Symptomatic[a,d,e]	0 (0)	1 (0.5)	0 (0)	4 (1.9)
SAE of ARIA-H[c]	0 (0)	0 (0)	0 (0)	0 (0)
Microhemorrhage[f]	41 (19.8)	36 (17.0)	41 (19.5)	41 (19.3)
Cortical superficial siderosis[f]	26 (12.6)	14 (6.6)	18 (8.6)	14 (6.6)
Macrohemorrhages[a,g]	1 (0.5)	2 (0.9)	0 (0)	1 (0.5)
SAE of macrohemmorrhage[c]	0 (0)	1 (0.5)	0 (0)	0 (0)
Cerebral hemorrhage[c]	0 (0)	1 (0.5)	0 (0)	0 (0)
Hemorrhagic stroke[c]	0 (0)	0 (0)	0 (0)	0 (0)
Any ARIA (either E or H)[a,b,d]	67 (32.4)	50 (23.6)	60 (28.6)	55 (25.8)
Any SAE of ARIA (either E or H)[c]	0 (0)	0 (0)	0 (0)	1 (0.5)
Concurrent ARIA-E and ARIA-H[f]	32 (15.5)	21 (9.9)	25 (11.9)	27 (12.7)

[a] Based on MRI or TEAE cluster

[b] ARIA-E TEAE cluster preferred terms are: amyloid-related imaging abnormalities edema/effusion, brain edema, and vasogenic cerebral edema

[c] Based on TEAE cluster

[d] ARIA-H TEAE cluster preferred terms are: ARIA-microhemorrhage and hemosiderin deposits, brainstem microhemorrhage, cerebellar microhemorrhage, cerebral hemosiderin deposit, cerebral microhemorrhage, and cortical superficial siderosis of the central nervous system

[e] Symptomatic ARIA-H low level term includes symptomatic ARIA-H, symptomatic ARIA-microhemorrhages and haemosiderin deposits, symptomatic ARIA-microhemorrhages and hemosiderin deposits, and symptomatic ARIA-cortical superficial siderosis

[f] Based on MRI only

[g] Macrohemorrhage preferred terms are cerebral hemorrhage and hemorrhagic stroke

Abbreviations: MRI, magnetic resonance imaging; N, number of participants in the analysis population; n, number of participants within each specific category; SAE, serious adverse event; SD, standard deviation; TEAE, treatment-emergent adverse event.

Table S2. ARIA relative risk reduction through 24 weeks

	Enhanced titration versus standard			Dose skipping versus standard			Cmax versus standard			
Category	Posterior RRR (Posterior SD)	95% CrI RRR	Posterior Probability of RRR ≥20%	Posterior RRR (Posterior SD)	95% CrI RRR	Posterior Probability of RRR ≥20%	Posterior RRR (Posterior SD)	95% CrI RRR	Posterior Probability of RRR ≥20%	Intercept prior was elicited as N
ARIA-E[a,b]	0.405 (0.123)	0.135, 0.616	**94.1**	0.195 (0.146)	-0.130, 0.447	51.2	0.211(0.145)	-0.097, 0.465	56.6	(-1.49, 8.10)
ARIA-H[a,c]	0.181 (0.145)	-0.138, 0.434	47.9	0.076 (0.156)	-0.268, 0.340	21.5	0.167 (0.145)	-0.145, 0.421	43.6	(-1.31, 8.55)
Microhemorrhage[d]	0.128 (0.175)	-0.262, 0.421	36.7	-0.003 (0.189	-0.424, 0.320	13.2	0.015 (0.185)	-0.393, 0.329	15.3	(-1.59, 7.88)
Cortical superficial siderosis[d]	0.450 (0.168)	0.066, 0.711	**92.3***	0.288 (0.197)	-0.163, 0.602	71.1	0.446 (0.167)	0.061, 0.709	**91.8***	(-2.34, 6.83)
Any ARIA (either E or H)[a,b,c]	0.261 (0.111)	0.023, 0.458	73.1	0.103 (0.130)	-0.175, 0.334	23.0	0.191 (0.120)	-0.066, 0.404	50.0	(-1.01, 9.55)
Concurrent ARIA-E and ARIA-H[d]	0.337 (0.168)	-0.043, 0.608	**80.5***	0.205 (0.185)	-0.206, 0.512	55.4	0.153 (0.194)	-0.278, 0.473	44.3	(-2.18, 9.99)

* Bold text indicates significant difference

[a] Based on MRI or TEAE cluster

[b] ARIA-E TEAE cluster preferred terms are: amyloid-related imaging abnormalities edema/effusion, brain edema, and vasogenic cerebral edema

[c] ARIA-H TEAE cluster preferred terms are: amyloid-related imaging abnormalities - microhemorrhage and hemosiderin deposits, brainstem microhemorrhage, cerebellar microhemorrhage, cerebral hemosiderin deposit, cerebral microhemorrhage, and cortical superficial siderosis of the central nervous system

[d] Based on MRI only

Abbreviations: CrI, credible interval; MRI, magnetic resonance imaging; RRR, relative risk reduction; TEAE, treatment-emergent adverse event.

Table S3. ARIA frequencies through 52 weeks

Category, n (%)	Standard (N=207)	Modified titration (N=212)	Dose skipping (N=210)	Cmax (N=213)
ARIA-E[a,b]	50 (24.2)	33 (15.6)	39 (18.6)	40 (18.8)
Asymptomatic[a,b]	40 (19.3)	27 (12.7)	28 (13.3)	29 (13.6)
Symptomatic[a,b]	10 (4.8)	6 (2.8)	11 (5.2)	11 (5.2)
SAE of ARIA-E[c]	0 (0)	1 (0.5)	4 (1.9)	2 (0.9)
ARIA-oedema/effusion[c]	0 (0)	1 (0.5)	4 (1.9)	2 (0.9)
Brain oedema[c]	0 (0)	0 (0)	0 (0)	0 (0)
Vasogenic cerebral oedema[c]	0 (0)	0 (0)	0 (0)	0 (0)
ARIA-H[a,d]	57 (27.5)	53 (25.0)	59 (28.1)	60 (28.2)
Asymptomatic[a,d]	57 (27.5)	52 (24.5)	58 (27.6)	57 (26.8)
Symptomatic[a,d,e]	0 (0)	1 (0.5)	1 (0.5)	3 (1.4)
SAE of ARIA-H[c]	0 (0)	0 (0)	0 (0)	0 (0)
Microhemorrhage[f]	44 (21.3)	48 (22.6)	52 (24.8)	53 (24.9)
Cortical superficial siderosis[f]	31 (15.0)	18 (8.5)	20 (9.5)	24 (11.3)
Macrohemorrhages[a,g]	1 (0.5)	2 (0.9)	1 (0.5)	1 (0.5)
SAE of macrohemmorrhage[c]	0 (0)	1 (0.5)	0 (0)	0 (0)
Cerebral hemorrhage[c]	0 (0)	1 (0.5)	0 (0)	0 (0)
Hemorrhagic stroke[c]	0 (0)	0 (0)	0 (0)	0 (0)
Any ARIA (either E or H)[a,b,d]	71 (34.3)	61 (28.8)	68 (32.4)	69 (32.4)
Any SAE of ARIA (either E or H)[c]	0 (0)	1 (0.5)	4 (1.9)	2 (0.9)
Concurrent ARIA-E and ARIA-H[f]	34 (16.4%)	23 (10.8%)	27 (12.9%)	30 (14.1%)

[a] Based on MRI or TEAE cluster

[b] ARIA-E TEAE cluster preferred terms are: amyloid-related imaging abnormalities oedema/effusion, brain oedema, and vasogenic cerebral oedema

[c] Based on TEAE cluster

[d] ARIA-H TEAE cluster preferred terms are: ARIA-microhemorrhage and hemosiderin deposits, brainstem microhemorrhage, cerebellar microhemorrhage, cerebral hemosiderin deposit, cerebral microhemorrhage, and cortical superficial siderosis of the central nervous system

[e] Symptomatic ARIA-H low level term includes symptomatic ARIA-H, symptomatic ARIA-microhemorrhages and haemosiderin deposits, symptomatic ARIA-microhemorrhages and hemosiderin deposits, and symptomatic ARIA-cortical superficial siderosis

[f] Based on MRI only

[g] Macrohemorrhage preferred term are cerebral hemorrhage and hemorrhagic stroke

Abbreviations: MRI, magnetic resonance imaging; N, number of participants in the analysis population; n, number of participants within each specific category; SAE, serious adverse event; SD, standard deviation; TEAE, treatment-emergent adverse event.

Table S4. ARIA relative risk reduction through 52 weeks

	Enhanced titration versus standard			Dose skipping versus standard			Cmax versus standard			
Category	Posterior RRR (Posterior SD)	95% CrI RRR	Posterior Probability of RRR ≥20%	Posterior RRR (Posterior SD)	95% CrI RRR	Posterior Probability of RRR ≥20%	Posterior RRR (Posterior SD)	95% CrI RRR	Posterior Probability of RRR ≥20%	Intercept prior was elicited as N
ARIA-E[a,b]	0.347 (0.129)	0.065, 0.569	**87.0***	0.218 (0.147)	-0.105, 0.472	58.4	0.209 (0.144)	-0.101, 0.455	55.7	(-1.42, 8.27)
ARIA-H[a,c]	0.087 (0.143)	-0.222, 0.340	22.2	-0.027 (0.152)	-0.357, 0.240	5.5	-0.034 (0.153)	-0.363, 0.238	4.7	(-1.06, 9.38)
Microhemorrhage[d]	-0.075 (0.190)	-0.496, 0.248	5.5	-0.168 (0.199)	-0.602, 0.174	1.6	-0.178 (0.199)	-0.616, 0.162	1.3	(-1.33, 8.50)
Cortical superficial siderosis[d]	0.412 (0.160)	0.047, 0.665	**90.0***	0.338 (0.169)	-0.043, 0.608	**80.9***	0.226 (0.181)	-0.177, 0.530	59.4	(-2.06, 7.12)
Any ARIA (either E or H)[a,b,c]	0.156 (0.116)	-0.087, 0.364	37.3	0.050 (0.126)	-0.215, 0.280	10.8	0.048 (0.123)	-0.214, 0.269	10.3	(-0.82, 10.40)
Concurrent ARIA-E and ARIA-H[d]	0.314 (0.166)	-0.058, 0.589	77.7	0.196 (0.180)	-0.203, 0.503	53.0	0.114 (0.195)	-0.329, 0.440	35.8	(-2.02, 7.18)

[*] Bold text indicates significant difference

[a] Based on MRI or TEAE cluster

[b] ARIA-E TEAE cluster preferred terms are: amyloid-related imaging abnormalities oedema/effusion, brain oedema, and vasogenic cerebral oedema

[c] ARIA-H TEAE cluster preferred terms are: amyloid-related imaging abnormalities - microhemorrhage and hemosiderin deposits, brainstem microhemorrhage, cerebellar microhemorrhage, cerebral hemosiderin deposit, cerebral microhemorrhage, and cortical superficial siderosis of the central nervous system

[d] Based on MRI only

Abbreviations: CrI, credible interval; MRI, magnetic resonance imaging; RRR, relative risk reduction; TEAE, treatment-emergent adverse event.

Table S5. ARIA-E relative risk reduction by *APOE* ε4 genotype through 24 weeks

	Modified titration versus standard			Dose skipping versus standard			Cmax versus standard		
APOE genotype	Posterior RRR (Posterior SD)	95% CrI RRR	Posterior probability of RRR ≥20%	Posterior RRR (Posterior SD)	95% CrI RRR	Posterior probability of RRR ≥20%	Posterior RRR (Posterior SD)	95% CrI RRR	Posterior probability of RRR ≥20%
Homozygote	0.362 (0.110)	0.133, 0.561	**92.4***	0.182 (0.118)	-0.072, 0.390	46.5	0.172 (0.117)	-0.075, 0.384	42.3
Heterozygote	0.407 (0.124)	0.138, 0.619	**94.2***	0.199 (0.150)	-0.133, 0.451	53.6	0.211 (0.145)	-0.102, 0.460	56.3
Noncarrier	0.449 (0.128)	0.167, 0.665	**96.1***	0.249 (0.162)	-0.116, 0.517	65.7	0.259 (0.156)	-0.084, 0.522	67.8

Note: ARIA-E by MRI or TEAE cluster. The posterior estimator of RRR is derived from a Bayesian multiple logistic regression model including donanemab dosing regimens, *APOE* ε4 genotype (heterozygote, homozygote, noncarrier), presence of microhemorrhage at baseline, presence of cortical superficial siderosis at baseline, and baseline amyloid terciles. The intercept prior was elicited as N (-1.42, 8.09).
* Bold text indicates significant difference.

Abbreviations: MRI, magnetic resonance imaging; RRR, relative risk reduction; SD, standard deviation; TEAE, treatment-emergent adverse event.

Table S6. ARIA-E relative risk reduction by *APOE* ε4 genotype through 52 weeks

	Modified titration versus standard			Dose skipping versus standard			Cmax versus standard		
APOE genotype	Posterior RRR (Posterior SD)	95% CrI RRR	Posterior probability of RRR ≥20%	Posterior RRR (Posterior SD)	95% CrI RRR	Posterior probability of RRR ≥20%	Posterior RRR (Posterior SD)	95% CrI RRR	Posterior probability of RRR ≥20%
Homozygote	0.294 (0.110)	0.06, 0.49	**80.8***	0.193 (0.114)	-0.04, 0.40	49.4	0.161 (0.115)	-0.08, 0.37	38.3
Heterozygote	0.340 (0.133)	0.05, 0.56	**85.5***	0.218 (0.146)	-0.10, 0.47	58.3	0.202 (0.145)	-0.11, 0.45	54.1
Noncarrier	0.382 (0.139)	0.07, 0.61	**89.9***	0.268 (0.157)	-0.08, 0.53	69.4	0.249 (0.156)	-0.10, 0.51	65.5

Note: ARIA-E by MRI or TEAE cluster. The posterior estimator of RRR is derived from a Bayesian multiple logistic regression model including donanemab dosing regimens, *APOE* ε4 genotype (heterozygote, homozygote, noncarrier), presence of microhemorrhage at baseline, presence of cortical superficial siderosis at baseline, baseline amyloid terciles. The intercept prior was elicited as N (-1.42, 8.27).
* Bold text indicates significant difference.

Abbreviations: MRI, magnetic resonance imaging; RRR, relative risk reduction; SD, standard deviation; TEAE, treatment-emergent adverse event.

Table S7. Safety overview through 24 weeks

Category[a] , n (%)	Standard (N=207)	Modified titration (N=212)	Dose skipping (N=210)	Cmax (N=213)
Deaths[b]	0	1 (0.5)	0	0
Serious AE	18 (8.7)	21 (9.9)	18 (8.6)	20 (9.4)
Discontinuations from study due to an AE	4 (1.9)	5 (2.4)	6 (2.9)	5 (2.3)
Discontinuations from study treatment due to an AE	8 (3.9)	11 (5.2)	8 (3.8)	14 (6.6)
TEAE	175 (84.5)	181 (85.4)	176 (83.8)	171 (80.3)
TEAE related to study treatment[c]	104 (50.2)	103 (48.6)	93 (44.3)	107 (50.2)

[a] Participants may be counted in more than one category.
[b] Deaths are also included as serious adverse events and discontinuations due to adverse events.
[c] Includes events that were considered related to study treatment as judged by the investigator.

Abbreviations: AE, adverse event; N, number of participants in the analysis population; n, number of participants with at least one AE per event type; TEAE, treatment-emergent AE.

Table S8. Safety overview through 52 weeks

Category[a] , n (%)	Standard (N=207)	Modified titration (N=212)	Dose skipping (N=210)	Cmax (N=213)
Deaths[b,c]	1 (0.5)	1 (0.5)	0 (0)	2 (0.9)
Serious AE	38 (18.4)	35 (16.5)	34 (16.2)	31 (14.6)
Discontinuations from study due to an AE	6 (2.9)	6 (2.8)	6 (2.9)	10 (4.7)
Discontinuations from study treatment due to an AE	14 (6.8)	15 (7.1)	19 (9.0)	19 (8.9)
TEAE	188 (90.8)	190 (89.6)	192 (91.4)	185 (86.9)
TEAE related to study treatment[d]	110 (53.1)	111 (52.4)	100 (47.6)	117 (54.9)

[a] Participants may be counted in more than one category.
[b] Deaths are also included as serious adverse events and discontinuations due to adverse events.
[c] Preferred term for cause of death: cardio-respiratory arrest (standard arm); cerebral haemorrhage (modified titration arm); pulmonary embolism (Cmax arm); myocardial infarction (Cmax arm)
[d] Includes events that were considered related to study treatment as judged by the investigator.

Abbreviations: AE, adverse event; N, number of participants in the analysis population; n, number of participants with at least one AE per event type; TEAE, treatment-emergent AE.

Table S9. Treatment-emergent adverse events ≥5.0% in any arm through week 24

Preferred term, n (%)	Standard (N=207)	Modified titration (N=212)	Dose skipping (N=210)	Cmax (N=213)
Participants with ≥1 TEAE	175 (84.5)	181 (85.4)	176 (83.8)	171 (80.3)
Amyloid-related imaging abnormality-oedema/effusion	49 (23.7)	29 (13.7)	38 (18.1)	39 (18.3)
Headache	41 (19.8)	32 (15.1)	44 (21.0)	34 (16.0)
Amyloid-related imaging abnormality-microhemorrhages and hemosiderin deposits	33 (15.9)	28 (13.2)	33 (15.7)	37 (17.4)
Infusion-related reaction	28 (13.5)	36 (17.0)	19 (9.0)	41 (19.2)
Fall	16 (7.7)	19 (9.0)	17 (8.1)	20 (9.4)
Dizziness	19 (9.2)	17 (8.0)	16 (7.6)	10 (4.7)
COVID-19	10 (4.8)	19 (9.0)	16 (7.6)	12 (5.6)
Urinary tract infection	7 (3.4)	16 (7.5)	18 (8.6)	9 (4.2)
Diarrhea	12 (5.8)	6 (2.8)	12 (5.7)	15 (7.0)
Fatigue	11 (5.3)	12 (5.7)	8 (3.8)	11 (5.2)
Upper respiratory tract infection	8 (3.9)	8 (3.8)	14 (6.7)	7 (3.3)
Cortical superficial siderosis of central nervous system	12 (5.8)	5 (2.4)	13 (6.2)	4 (1.9)
Arthralgia	8 (3.9)	13 (6.1)	5 (2.4)	4 (1.9)

Abbreviations: AE, adverse event; COVID-19: Coronavirus Disease 2019; N, number of participants in the analysis population; n, number of participants within each specific category; TEAE, treatment-emergent adverse event

Table S10. Treatment-emergent adverse events ≥5.0% in any arm through week 52

Preferred term, n (%)	Standard (N=207)	Modified titration (N=212)	Dose skipping (N=210)	Cmax (N=213)
Participants with ≥1 TEAE	188 (90.8)	190 (89.6)	192 (91.4)	185 (86.9)
Headache	44 (21.3)	39 (18.4)	49 (23.3)	40 (18.8)
Amyloid-related imaging abnormality-microhaemorrhages and haemosiderin deposits	40 (19.3)	36 (17.0)	41 (19.5)	49 (23.0)
Amyloid-related imaging abnormality-oedema/effusion	50 (24.2)	33 (15.6)	38 (18.1)	40 (18.8)
Infusion-related reaction	29 (14.0)	37 (17.5)	19 (9.0)	42 (19.7)
Fall	25 (12.1)	29 (13.7)	29 (13.8)	34 (16.0)
COVID-19	18 (8.7)	28 (13.2)	35 (16.7)	19 (8.9)
Dizziness	29 (14.0)	22 (10.4)	17 (8.1)	13 (6.1)
Urinary tract infection	13 (6.3)	17 (8.0)	24 (11.4)	12 (5.6)
Upper respiratory tract infection	13 (6.3)	16 (7.5)	20 (9.5)	14 (6.6)
Diarrhoea	14 (6.8)	12 (5.7)	16 (7.6)	18 (8.5)
Fatigue	15 (7.2)	12 (5.7)	11 (5.2)	14 (6.6)
Arthralgia	13 (6.3)	17 (8.0)	11 (5.2)	11 (5.2)
Back pain	10 (4.8)	12 (5.7)	12 (5.7)	16 (7.5)
Cortical superficial siderosis of central nervous system	16 (7.7)	9 (4.2)	13 (6.2)	7 (3.3)
Nausea	9 (4.3)	12 (5.7)	8 (3.8)	12 (5.6)
Cough	5 (2.4)	8 (3.8)	13 (6.2)	11 (5.2)
Vomiting	5 (2.4)	8 (3.8)	6 (2.9)	11 (5.2)
Sinusitis	11 (5.3)	7 (3.3)	6 (2.9)	4 (1.9)
Insomnia	10 (4.8)	7 (3.3)	6 (2.9)	3 (1.4)
Contusion	12 (5.8)	6 (2.8)	4 (1.9)	1 (0.5)

Abbreviations: AE, adverse event; COVID-19, Coronavirus Disease 2019; N, number of participants in the analysis population; n, number of participants within each specific category; TEAE, treatment-emergent adverse event

Table S11. Planned and observed cumulative dose and exposure metrics across study arms

Parameter	Standard (N=206)	Modified titration (N=212)	Dose skipping (N=210)	Cmax (N=213)
Cumulative dose amount (0–16 weeks)				
Planned (mg)	3500	3500	3500	3500
Observed (mg)	3123	3015	3183	2990
Mean observed $AUC_{(0–12\ weeks)}$, µg*h/mL (25th–75th percentile)	58700 (49400–69580)	49300 (40557–66313)	NA	NA
Area under the concentration versus time curve during 1 dosing interval at steady state				
Cav,ss µg/ml (95%CI)	63.1 (58.1–65.9)	63.0 (58.0–65.9)	NA	NA
Cav,ss GMR (95% CI)	0.998 (0.973, 1.029)		NA	

Notes: GMR (95%CI) were generated using sampling from the bootstrapped CI of the post hoc population PK parameters, similarly to Wang et al., 2024 [14]. Comparisons only reported for standard and modified titration arms. PK non-inferiority for modified titration arm is demonstrated with lower bound of 95% CI for GMR≥0.8, Wang et al., 2024 [14].

Abbreviations: AUC, area under the curve; Cav,ss, average concentration at steady state; CI, confidence interval around the mean; GMR, geometric mean ratio; N, number of participants in each arm; NA, not applicable

Table S12. Amyloid assessments through 24 weeks

Parameter	Standard (N=194)	Modified titration (N=201)	Dose skipping (N=199)	Cmax (N=199)
Adjusted mean change from baseline*, CL (SE)	-58.8 (1.8)	-56.3 (1.7)	-58.7 (1.7)	-51.0 (1.7)
Amyloid below 24.1 CL at 24 weeks, n (%)	110 (56.7)	102 (50.7)	120 (60.3)	97 (48.7)
Dose stopping criteria met (<11 CL), n (%)	66 (34.0)	65 (32.3)	78 (39.2)	60 (30.2)

* The ANCOVA model is: postbaseline amyloid Centiloid = baseline amyloid Centiloid + dosing regimen + baseline age.

Abbreviations: ANCOVA, analysis of covariance; CL, Centiloid; SE, standard error; n, number of participants within each specific category.

Table S13. Amyloid assessments through 52 weeks

Parameter	Standard (N=173)	Modified titration (N=178)	Dose skipping (N=186)	Cmax (N=174)
Adjusted mean change from baseline*, CL (SE)	-71.2 (1.6)	-70.3 (1.6)	-68.4 (1.6)	-66.6 (1.6)
Amyloid below 24.1 CL at 52 weeks, n (%)	138 (79.8)	137 (77.0)	151 (81.2)	130 (74.7)
Dose stopping criteria met, n (%)	116 (67.1)	117 (65.7)	135 (72.6)	101 (58.0)

*Adjusted mean change from baseline and SE are derived using a mixed model for repeated measures methodology with fixed factors being treatment, visit, treatment-by-visit interaction, baseline PET value, baseline PET value-by-visit interaction, and baseline age. Variance-covariance structure is unstructured.

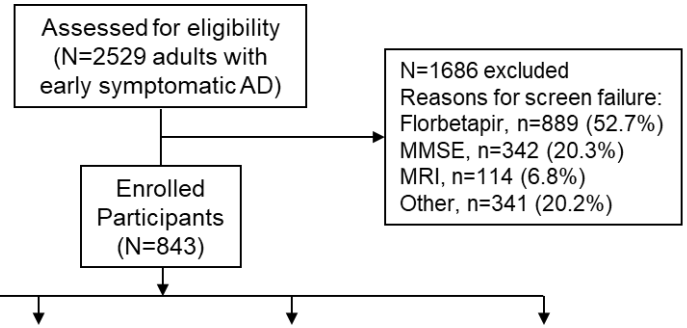
Abbreviations: CL, Centiloid; n, number of participants within each specific category; PET. positron emission tomography; SE, standard error.

Figure S1. Participant disposition

(A) study disposition and (B) treatment disposition. The enrolled population (N = 843) consisted of all participants who passed screening and were enrolled to receive donanemab.
*Visit 12 date was used on disposition form.

Abbreviations: AD, Alzheimer's disease; MMSE, Mini-Mental State Examination; MRI, magnetic resonance imaging; N, number of participants in the analysis population; n, number of participants within each specific category.

(A) Study disposition



	Standard (N=208)		Modified Titration (N=212)		Dose Skipping (N=210)		Cmax (N=213)		Total (N=843)	
	Through 24-weeks	Through 52-weeks	Through 24-weeks	Through 52-weeks	Through 24-weeks	Through 52-weeks	Through 24-weeks	Through 52-weeks	Through 24-weeks	Through 52-weeks
Discontinued Study	14 (6.7)	29 (13.9)	19 (9.0)	29 (13.7)	14 (6.7)	21 (10.0)	22 (10.3)	32 (15.0)	69 (8.2)	111 (13.2)
Withdrawal by participant, n (%)	8 (3.8)	13 (6.3)	7 (3.3)	12 (5.7)	5 (2.4)	11 (5.2)	13 (6.1)	15 (7.0)	33 (3.9)	51 (6.0)
Adverse event, n (%)	4 (1.9)	5 (2.4)	4 (1.9)	5 (2.4)	6 (2.9)	6 (2.9)	5 (2.3)	8 (3.8)	19 (2.3)	24 (2.8)
Physician decision, n (%)	0 (0)	2 (1.0)	4 (1.9)	4 (1.9)	2 (1.0)	2 (1.0)	1 (0.5)	1 (0.5)	7 (0.8)	9 (1.1)
Lost to follow-up, n (%)	1 (0.5)	6 (2.9)	2 (0.9)	3 (1.4)	1 (0.5)	2 (1.0)	1 (0.5)	1 (0.5)	5 (0.6)	12 (1.4)
Other, n (%)	0	1 (0.5)	1 (0.5)	1 (0.5)	0	0	1 (0.5)	1 (0.5)	2 (0.2)	3 (0.4)
Assigned treatment by mistake, n (%)	0	0	0	0	0	0	1 (0.5)	1 (0.5)	1 (0.1)	1 (0.1)
Death, n (%)	0	1 (0.5)	1 (0.5)	1 (0.5)	0	0	0	2 (0.9)	1 (0.1)	4 (0.5)
Withdrawal due to caregiver circumstances, n (%)	1 (0.5)	1 (0.5)	0	0	0	0	0	1 (0.5)	1 (0.1)	2 (0.2)
Continuing Study, n (%)	194 (93.3)	179 (86.1)	193 (91.0)	183 (86.3)	196 (93.3)	189 (90.0)	191 (89.7)	181 (85.0)	774 (91.8)	732 (86.8)

(B) Treatment disposition

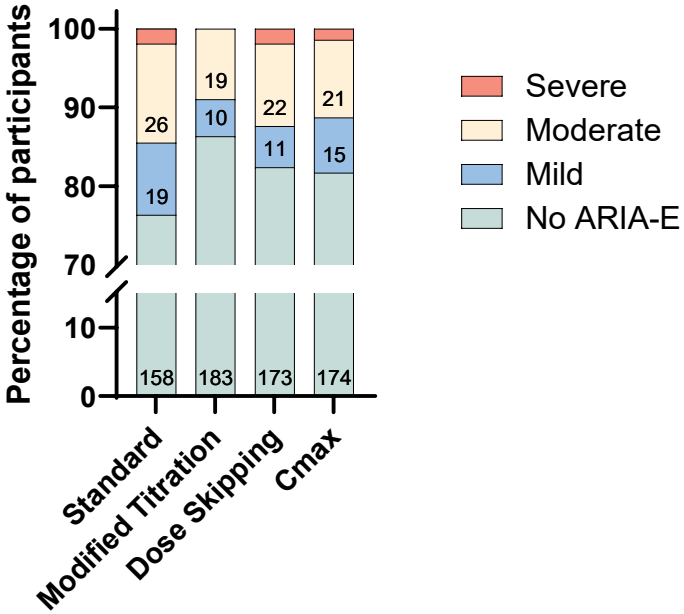
	Standard (N=208)		Modified Titration (N=212)		Dose Skipping (N=210)		Cmax (N=213)		Total (N=843)	
	Through 24-weeks	Through 52-weeks	Through 24-weeks	Through 52-weeks	Through 24-weeks	Through 52-weeks	Through 24-weeks	Through 52-weeks	Through 24-weeks	Through 52-weeks
Discontinued treatment	19 (9.1)	103 (49.5)	28 (13.2)	105 (49.5)	18 (8.6)	114 (54.3)	37 (17.4)	104 (48.8)	102 (12.1)	426 (50.5)
Withdrawal by participant, n (%)	7 (3.4)	12 (5.8)	7 (3.3)	12 (5.7)	5 (2.4)	13 (6.2)	13 (6.1)	13 (6.1)	32 (3.8)	50 (5.9)
Adverse event, n (%)	8 (3.8)	14 (6.7)	11 (5.2)	14 (6.6)	9 (4.3)	19 (9.0)	14 (6.6)	18 (8.5)	42 (5.0)	65 (7.7)
Dose stopping criteria met*, n (%)	1 (0.5)	67 (32.2)	2 (0.9)	65 (30.7)	1 (0.5)	78 (37.1)	5 (2.3)	63 (29.6)	9 (1.1)	273 (32.4)
Physician decision, n (%)	1 (0.5)	2 (1.0)	4 (1.9)	4 (1.9)	2 (1.0)	2 (1.0)	1 (0.5)	4 (1.9)	8 (0.9)	12 (1.4)
Lost to follow-up, n (%)	0	4 (1.9)	2 (0.9)	3 (1.4)	1 (0.5)	2 (1.0)	1 (0.5)	1 (0.5)	4 (0.5)	10 (1.2)
Other, n (%)	1 (0.5)	2 (1.0)	1 (0.5)	4 (1.9)	0	0	1 (0.5)	2 (0.9)	3 (0.4)	8 (0.9)
Assigned treatment by mistake, n (%)	0	1 (0.5)	0	0	0	0	1 (0.5)	1 (1.5)	1 (0.1)	2 (0.2)
Death, n (%)	0	0	1 (0.5)	1 (0.5)	0	0	0	1 (0.5)	1 (0.1)	2 (0.2)
Progressive disease, n (%)	0	0	0	2 (0.9)	0	0	1 (0.5)	0	1 (0.1)	2 (0.2)
Withdrawal due to caregiver circumstances, n (%)	1 (0.5)	1 (0.5)	0	0	0	0	0	1 (0.5)	1 (0.1)	2 (0.2)
Continuing Treatment, n (%)	189 (90.9)	105 (50.5)	184 (86.8)	107 (50.5)	192 (91.4)	96 (45.7)	176 (82.6)	109 (51.2)	741 (87.9)	417 (49.5)

Figure S2. ARIA-E severity and time to first incidence

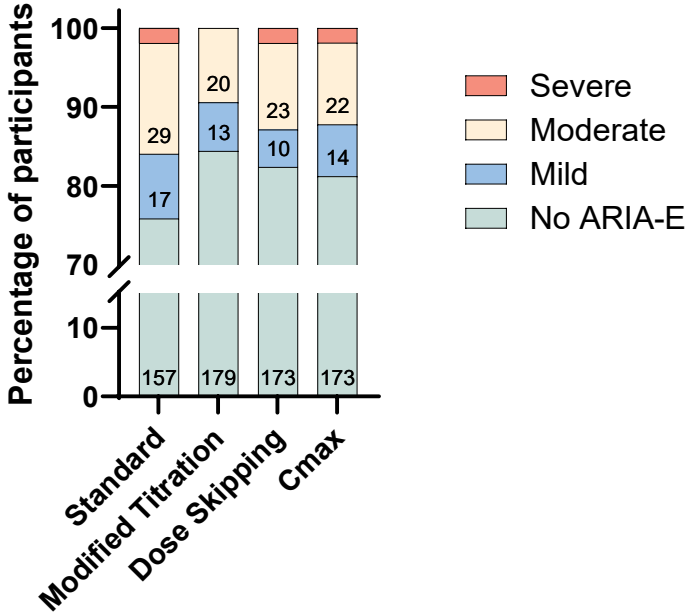
ARIA-E maximum radiographic severity based on Cochran-Mantel-Haenszel test through (A) 24 weeks and (B) 52 weeks. (C) Kaplan-Meier curve showing hazard of time to first ARIA-E through 52 weeks, based on MRI only. The percentage of participants with ARIA-E is calculated using the number of participants who experienced ARIA-E by the given timepoint as the numerator and the number of participants still active in the study as the denominator. Participants at risk are the number of participants still active in the study and who have not yet experienced ARIA-E.

Abbreviation: MRI, magnetic resonance imaging.

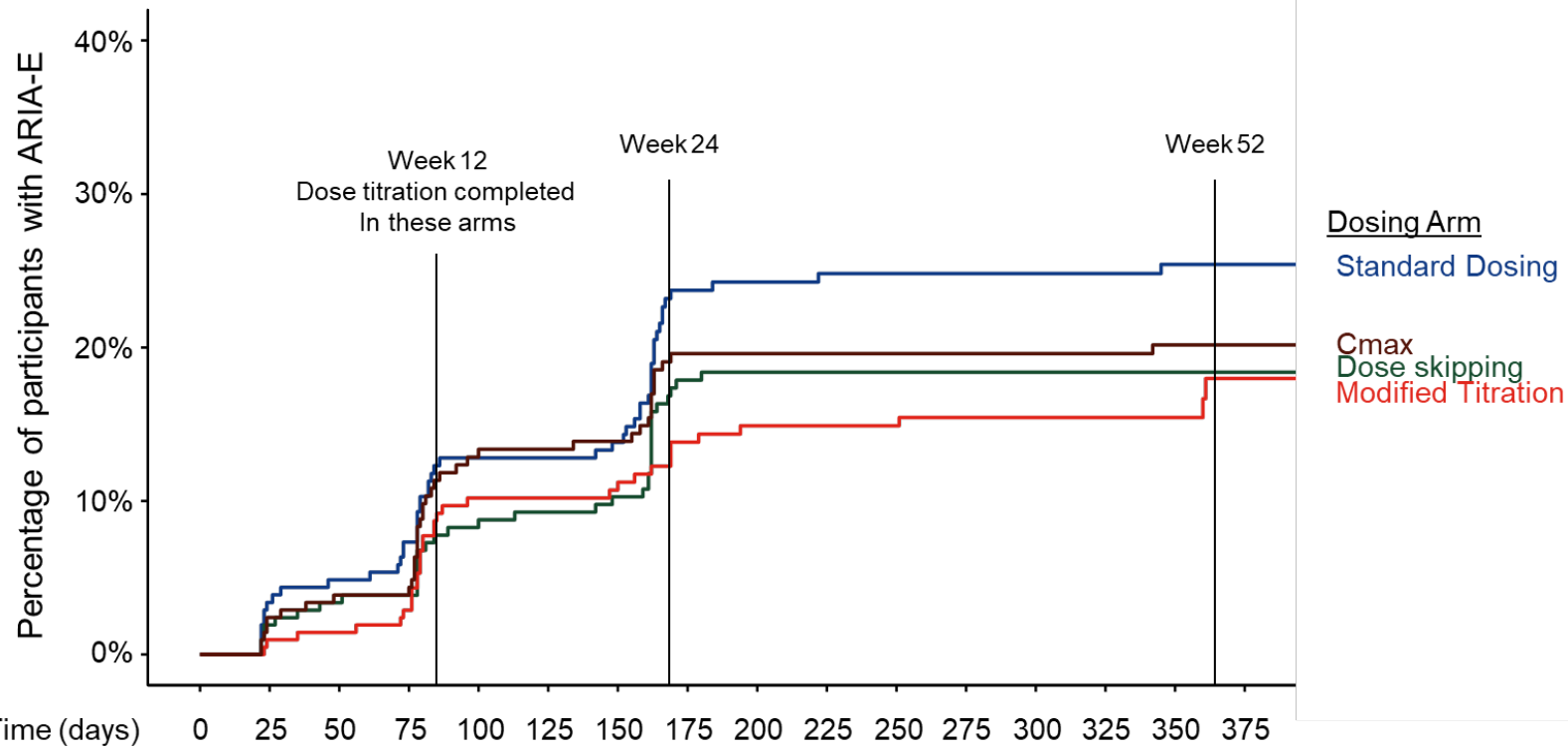
(A) ARIA-E severity through 24 weeks



(B) ARIA-E severity through 52 weeks



(C) Cumulative hazard of time to first ARIA-E through 52 weeks



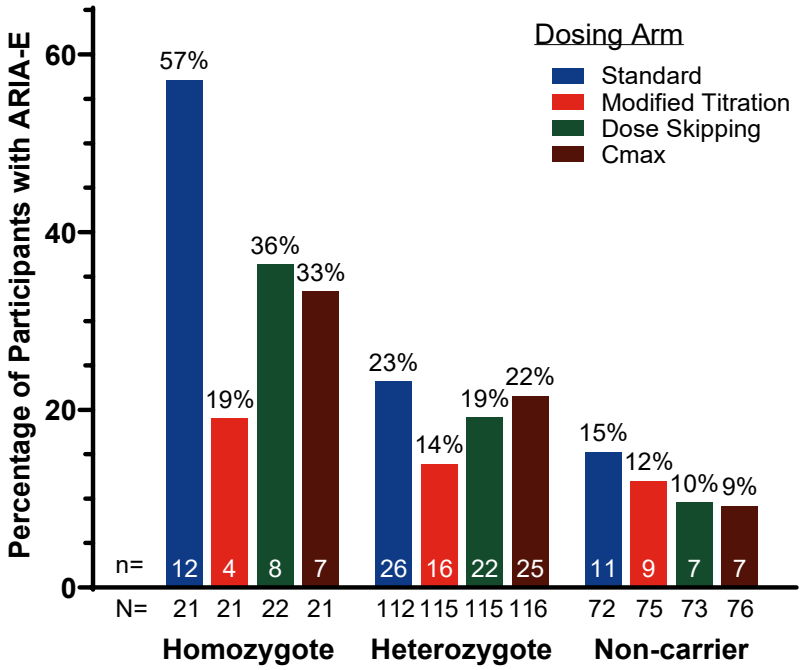
Time (days)	0	25	50	75	100	125	150	175	200	225	250	275	300	325	350	375
Number of participants at risk																
Standard	207	198	193	188	172	171	169	141	138	136	136	135	134	133	123	15
Modified Titration	212	207	205	201	179	174	173	164	160	159	159	157	156	154	147	16
Dose Skipping	210	203	199	197	184	181	178	160	159	158	158	157	157	156	145	15
Cmax	213	201	196	194	171	169	167	149	149	149	148	148	148	148	135	8

Figure S3. ARIA-E by *APOE* ε4 genotype

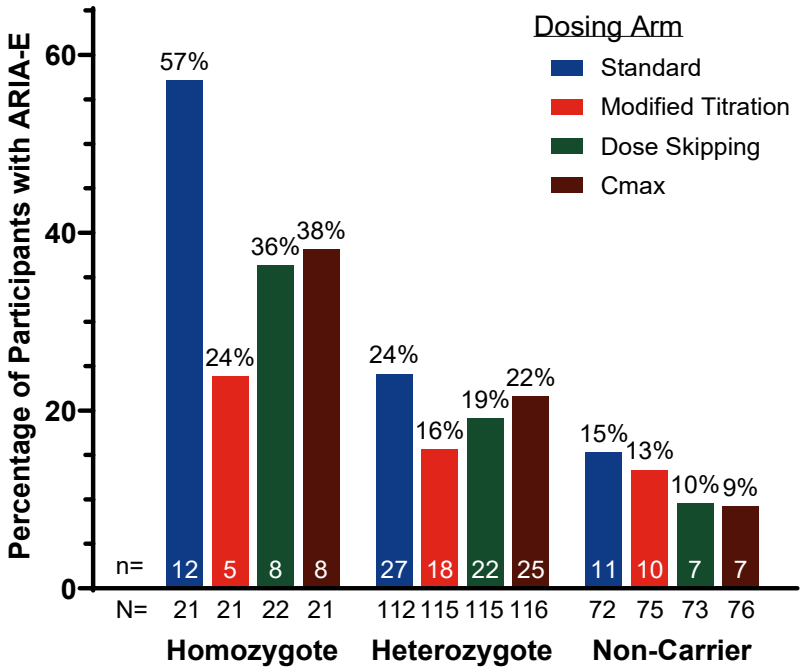
Based on safety MRI, bar graphs show the frequency of ARIA-E by *APOE* ε4 genotype through (A) 24 weeks and (B) 52 weeks and the frequency of symptomatic ARIA-E by *APOE* ε4 genotype through (C) 24 weeks and (D) 52 weeks.

Abbreviations: *APOE,* apolipoprotein E; MRI, magnetic resonance imaging; N, number of participants in the analysis population; n, number of participants within each specific category.

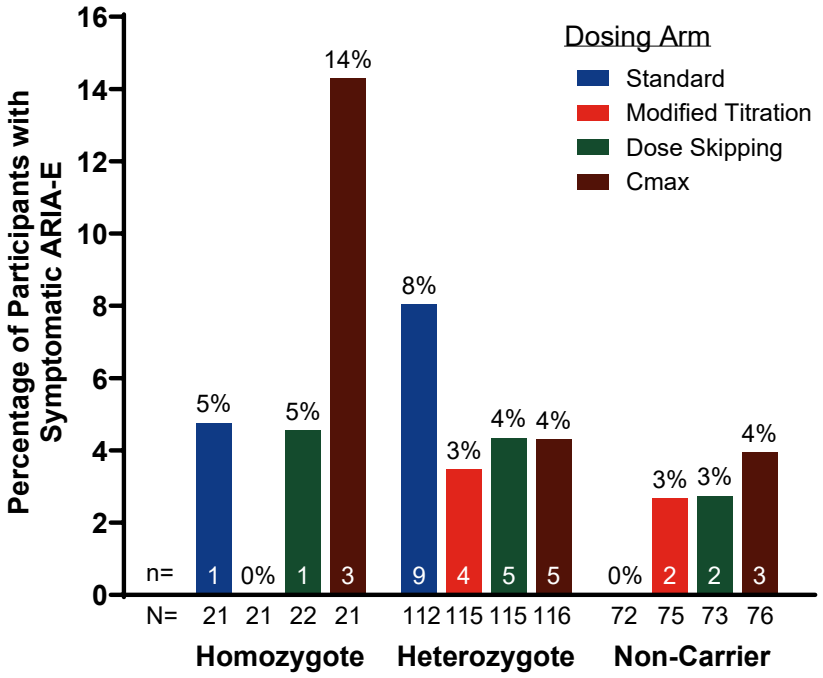
(A) ARIA-E through 24 weeks



(B) ARIA-E through 52 weeks



(C) Symptomatic ARIA-E through 24 weeks



(D) Symptomatic ARIA-E through 52 weeks

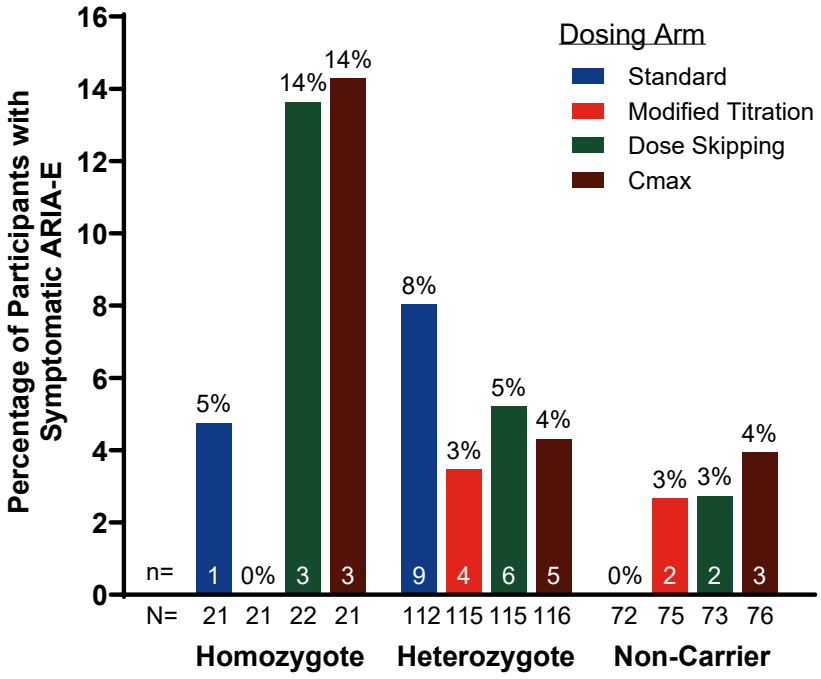
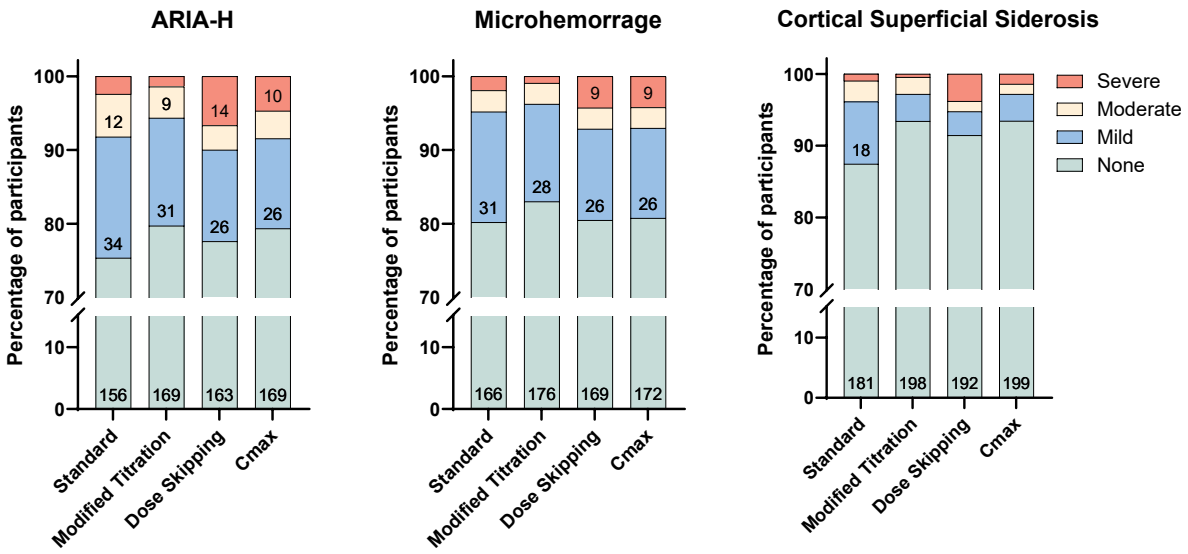


Figure S4. Severities of ARIA H, microhemorrage, and cortical superficial siderosis

Maximum radiographic severity through week 24 (A) and week 52 (B). Numbers shown are the number of participants who experienced the event at the given severity level. Denominator is the number of participants who experienced the event.

(A) ARIA-H severities through week 24



(B) ARIA-H severities through week 52

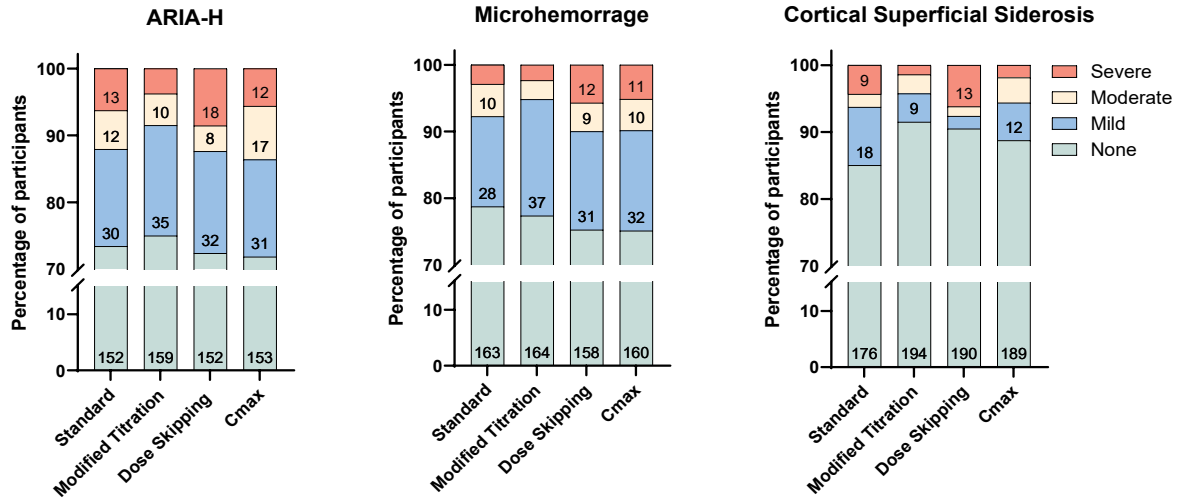
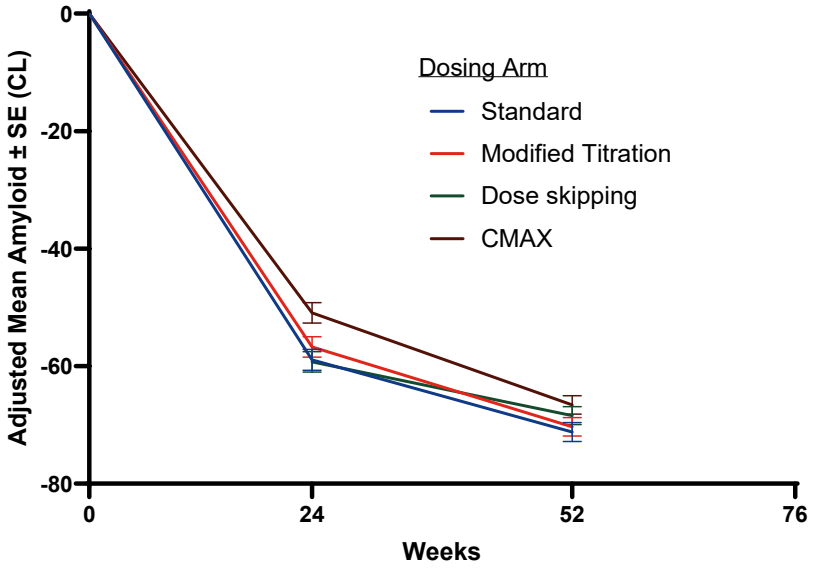
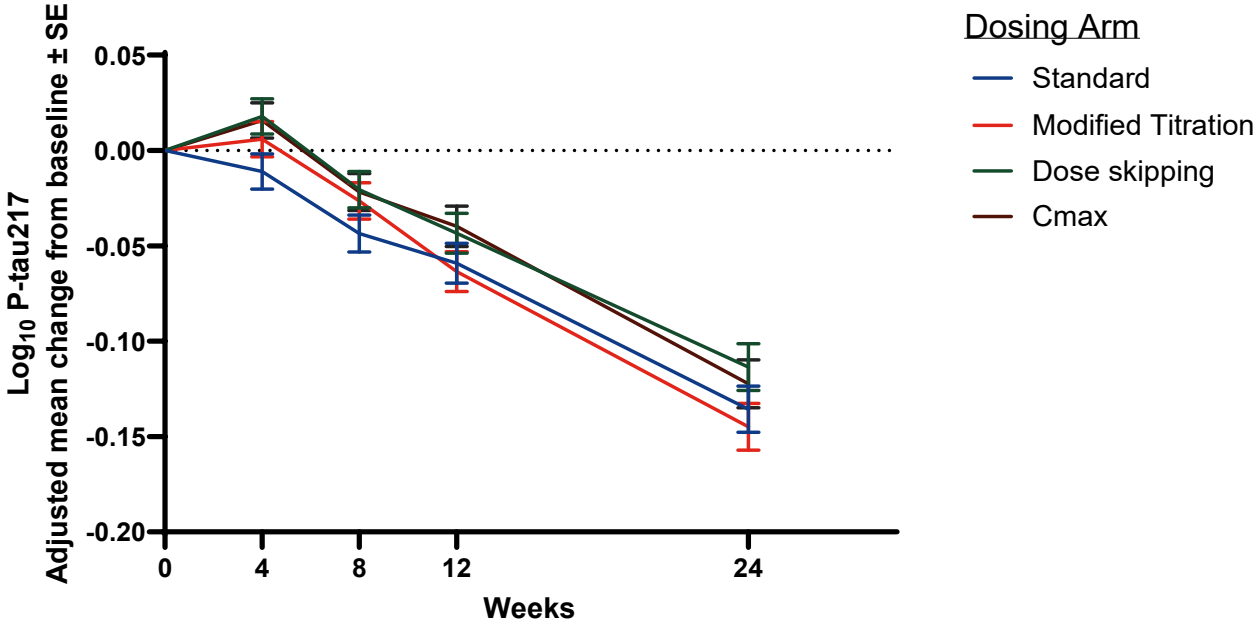


Figure S5. Biomarkers

(A) Amyloid levels (reduction) through 52 weeks. Adjusted mean change from baseline and SE are derived using MMRM methodology with fixed factors being treatment, visit, treatment-by-visit interaction, baseline PET value, baseline PET value-by-visit interaction, and baseline age with an unstructured variance-covariance. (B) Plasma P-tau217 through 24 weeks as assessed on Eli Lilly and Company's Meso scale discovery platform using MMRM with fixed factors being treatment, visit, treatment-by-visit interaction, baseline score, baseline score-by-visit interaction, and age at baseline with an unstructured variance-covariance.

Abbreviations: CL, Centiloid; MMRM, mixed model for repeated measures PET, positron emission tomography; SE, standard error.

(A) Change in amyloid through 52 weeks



(B) Change in plasma P-tau217 through 24 weeks